Design and Implementation of an FSM-Based AMBA AXI Bus Arbiter Using Verilog HDL

Group 13

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Introduction

- AXI (Advanced eXtensible Interface) is a widely used bus protocol in modern SoC and FPGA systems.
- Efficient data transfer between multiple masters and slaves requires proper arbitration.
- Our project focuses on designing FSM-based arbiters for AXI read and write channels.
- Importance: Used in CPUs, GPUs, and real-time embedded systems to ensure fairness and throughput.

Motivation

- AXI interconnects are widely used in industry (ARM-based SoCs, FPGA IP cores).
- Arbitration is the backbone of multi-master communication, but often treated as a "black box".
- Implementing arbiter logic from scratch provides deep protocol understanding.
- Expected Contribution:
  - A working FSM-based AXI arbiter (read + write).
  - Verified through simulation in Vivado.

Problem Statement

- Modern SoCs have multiple masters (CPU, DMA, peripherals) needing simultaneous memory access.
- Without efficient arbitration, issues like starvation, unfair access, and protocol violation occur.
- Existing arbiters (fixed-priority) lack flexibility and fairness.
- Our work: FSM-based arbiter supporting multiple masters with fair request handling.

Objectives

- Design an FSM-based arbiter for AXI read and write channels.
- Implement in Verilog HDL and simulate using Xilinx Vivado.
- Ensure fairness and avoid starvation in multi-master environments.
- Document and analyze performance using testbenches and waveform analysis.

Literature Review

Sl No	Authors	Year	Title	Inference
[1]	R. Khanam and Z. Ahmad	2016	Finite state machine based arbiter for on chip communication architecture	1. Proposes FSM-based arbitration for on-chip bus systems. 2. Reduces bus contention and improves predictability.
[2]	R. Bhaktavatchalu et al.	2016	Design of AXI bus interface modules on FPGA	1. Presents FPGA design of AXI bus modules. 2. Focuses on modular and reusable AXI interfaces. 3. Demonstrates efficient communication between masters and slaves.
[3]	F. Restuccia et al.	2019	Is Your Bus Arbiter Really Fair? Restoring Fairness in AXI Interconnects for FPGA SoCs	1. Evaluates fairness issues in AXI arbiters. 2. Introduces fairness-restoring mechanisms for arbitration.
[4]	J. Luo et al.	2022	A Low-Latency Fair-Arbiter Architecture for Network-on-Chip Switches	1. Proposes low-latency arbiter for NoC-based SoCs. 2. Ensures fairness while minimizing response delay. 3. Improves throughput in multi-core

Literature Review Contd.

Sl No	Authors	Year	Title	Inference
[5]	T. Benz et al.	2023	AXI-REALM: A Lightweight and Modular Interconnect Extension for Traffic Regulation and Monitoring of Heterogeneous Real-Time SoCs	1. Extends AXI protocol with monitoring and regulation. 2. Provides lightweight modular IP for SoCs. 3. Supports traffic control in real-time heterogeneous environments.
[6]	K. A. Sangamnerkar et al.	2023	Scalable Design and Implementation IP for Advanced Extensible Interface (AXI) Protocol	1. Presents scalable AXI IP architecture. 2. Ensures reusability and efficient hardware design. 3. Improves system scalability in large SoCs.
[7]	L. S. B et al.	2024	Design and Verification of AMBA AXI Protocol	1. Implements AXI protocol verification on FPGA. 2. Focuses on correctness and timing validation. 3. Demonstrates simulation and hardware testing results.

Proposed Methodology

- The design implements an **FSM-based AXI Arbiter** that controls read and write transactions between multiple masters and a shared slave.
- States include: **Idle, Request Capture, Arbitration, Grant, Data Transfer, Response Handling**.
- Arbitration techniques used: **Round Robin** to ensure fairness and reduce bus contention.
- Parallel arbitration with address decoding minimizes latency and improves throughput.
- AXI handshaking signals (VALID/READY) integrated for reliable communication.

Block Diagram

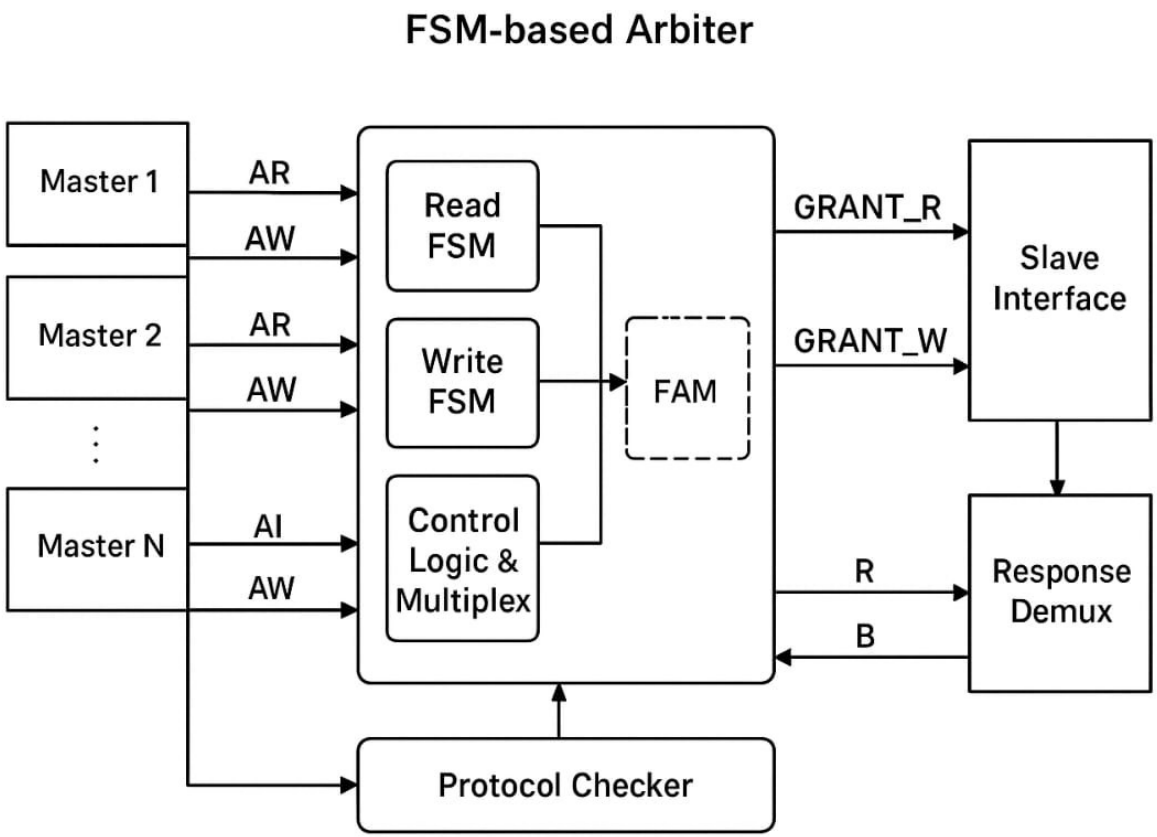
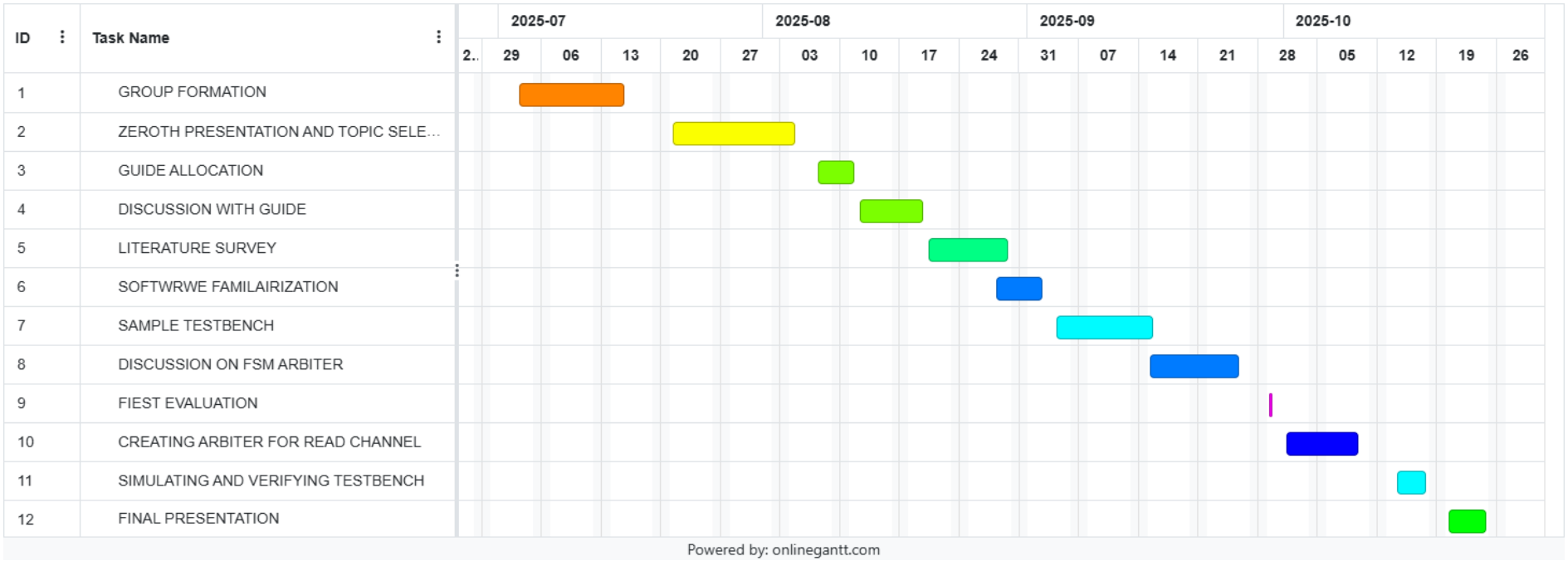


Figure: FSM Based Arbiter For AMBA AXI BUS Protocol

Project Planning

Gantt Chart:



Resource Identification

Hardware Requirements:

- Xilinx FPGA Board (e.g., Nexys A7)

Software Requirements:

- Xilinx Vivado

Task Allocation:

- Member 1 – Literature review on AXI,Arbitration Techniques.
- Member 2 – FSM Design(Read/Write)
- Member 3 – Verilog coding Testbench
- Member 4 - Simulation ,Verification

Current Status

- Collected and reviewed literature on AMBA and AXI bus arbitration methods.
- Understood AXI protocol structure and identified the need of arbitration.
- Familiarized with the Vivado software.
- Wrote the preliminary Verilog design for the arbiter module.
- Created and compiled a basic testbench structure.
- Verified design syntax correctness in Vivado environment

Conclusion

- FSM-based arbitration ensures fair, predictable, and efficient bus access among multiple masters in AXI interconnects.
- The proposed arbiter design demonstrates low latency and improved throughput compared to conventional arbitration methods.
- FPGA implementation validates the practicality, correctness, and timing efficiency of the design.
- The literature shows gaps in scalability, fairness, and real-time support, which this work addresses.
- The design contributes towards robust and high-performance SoC communication architectures suitable for next-generation systems.

References

[1] R. Khanam and Z. Ahmad, "Finite state machine based arbiter for on chip communication architecture," *2016 International Conference on Computing, Communication and Automation (ICCCA)*, Greater Noida, India, 2016.

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[5] T. Benz, A. R. Golla, H. Y. T. Nyuyen, L. Bauer and T. Wild, "AXI-REALM: A Lightweight and Modular Interconnect Extension for Traffic Regulation and Monitoring of Heterogeneous Real-Time SoCs," *arXiv preprint* arXiv:2311.09662, 2023.

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Thank You!